

NYURBA, Russian Federation

WMO: 246390

Lat: 63.300N

Lon: 118.353E

Elev: 119

StdP: 99.90

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-50.7	-47.7	-52.3	0.0	-50.5	-49.7	0.0	-47.7	7.6	-20.6	6.4	-21.3	0.4	230	0.722

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	29.5	19.0	27.3	17.9	25.1	16.9	20.2	27.1	19.0	25.3	17.9	23.6	2.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.8	12.9	22.9	16.5	11.9	21.8	15.4	11.1	20.7	58.6	27.1	54.6	25.3	51.0	23.6	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.2	6.0	5.2	-51.1	32.7	3.4	1.8	-53.6	34.0	-55.6	35.1	-57.5	36.1	-60.0	37.5		
			-51.2	22.3	3.4	1.5	-53.7	23.4	-55.6	24.2	-57.5	25.1	-60.0	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-7.7	-33.9	-29.7	-18.3	-4.9	6.0	15.0	17.5	13.9	5.3	-6.7	-24.1	-33.5
	DBStd	19.92	8.67	8.16	8.40	6.27	4.67	4.63	3.88	3.72	4.39	7.01	9.25	9.02
	HDD10.0	7004	1361	1112	877	446	141	11	1	10	153	519	1024	1349
	HDD18.3	9554	1619	1346	1135	696	383	118	62	144	391	777	1274	1607
	CDD10.0	553	0	0	0	0	17	161	234	129	12	0	0	0
	CDD18.3	60	0	0	0	0	0	18	37	6	0	0	0	0
	CDH23.3	863	0	0	0	0	7	277	478	101	1	0	0	0
	CDH26.7	241	0	0	0	0	0	74	148	19	0	0	0	0
Wind	WSAvg	2.2	1.9	1.8	1.9	2.4	2.9	2.5	2.1	2.1	2.3	2.4	2.0	1.8
Precipitation	PrecAvg	288	12	9	8	11	25	43	43	51	30	25	17	14
	PrecMax	378	23	26	21	31	55	100	96	95	69	42	38	39
	PrecMin	198	6	2	2	0	2	15	7	6	2	7	4	3
	PrecStd	45	5	5	4	8	13	21	21	24	16	9	7	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-9.0	-5.4	4.8	13.6	24.2	31.4	32.8	29.2	21.3	7.6	0.4	-6.0
		MCWB	-9.5	-6.1	0.7	6.6	13.0	18.6	20.2	18.7	13.2	4.4	-0.7	-6.3
	2%	DB	-14.0	-10.2	0.7	9.8	20.6	28.8	30.3	26.0	18.1	5.2	-2.1	-12.4
		MCWB	-14.6	-10.9	-2.0	4.3	11.2	18.1	19.7	17.0	11.5	2.9	-2.6	-12.7
	5%	DB	-17.6	-13.8	-2.3	7.0	17.7	26.7	28.2	23.9	15.1	3.4	-5.5	-16.6
		MCWB	-18.0	-14.3	-4.2	2.5	9.7	16.9	18.8	16.1	10.1	1.6	-6.0	-16.9
	10%	DB	-21.2	-17.4	-4.9	4.7	15.0	24.3	26.0	21.7	12.6	1.7	-9.8	-20.5
		MCWB	-21.4	-17.8	-6.5	1.0	8.4	15.9	18.0	15.4	8.7	0.3	-10.3	-20.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-9.5	-6.0	1.1	7.0	14.0	21.4	22.2	20.1	14.2	5.1	-0.3	-6.3
		MCDB	-9.0	-5.5	4.6	13.0	22.3	27.6	28.9	27.2	19.6	6.8	0.2	-6.0
	2%	WB	-14.4	-10.8	-1.9	4.6	11.7	19.3	20.7	18.4	12.5	3.2	-2.8	-12.7
		MCDB	-14.0	-10.2	0.5	9.4	19.1	26.1	27.7	23.7	16.6	4.8	-2.0	-12.4
	5%	WB	-18.0	-14.4	-4.2	2.8	10.2	18.0	19.7	17.2	10.7	1.8	-6.0	-16.9
		MCDB	-17.6	-13.9	-2.4	6.4	17.1	24.8	26.5	22.1	14.0	3.1	-5.5	-16.6
	10%	WB	-21.5	-17.8	-6.5	1.3	8.6	16.7	18.6	16.0	9.2	0.4	-10.2	-20.7
		MCDB	-21.2	-17.4	-5.0	4.4	14.5	22.9	24.6	20.7	12.2	1.7	-9.7	-20.5
Mean Daily Temperature Range	5% DB	MDBR	7.7	10.3	14.7	12.9	10.5	12.1	11.8	11.6	9.4	7.2	8.2	7.2
		MCDBR	9.4	11.6	12.9	13.7	15.5	16.2	16.3	16.1	13.9	7.6	8.9	10.2
		MCWBR	9.2	11.2	10.9	9.2	8.3	7.2	7.3	8.2	7.9	5.8	8.7	10.1
	5% WB	MCDBR	9.5	11.6	12.5	12.1	14.2	14.2	14.0	13.0	12.0	6.6	8.9	10.2
		MCWBR	9.3	11.2	10.5	8.3	8.1	7.1	7.0	7.3	7.2	5.5	8.8	10.0
Clear-Sky Solar Irradiance	taub	0.212	0.257	0.293	0.332	0.365	0.410	0.472	0.448	0.326	0.277	0.207	0.191	
	taud	1.917	2.063	2.089	2.116	2.310	2.285	2.081	2.101	2.463	2.329	1.951	1.880	
	Ebn at Noon	384	667	792	839	846	808	733	709	774	660	384	193	
	Edh at Noon	33	67	99	122	111	117	140	125	71	52	30	16	
All-Sky Solar Radiation	RadAvg	0.26	1.19	2.93	4.94	5.44	5.71	5.27	3.94	2.43	1.06	0.39	0.11	
	RadStd	0.02	0.06	0.10	0.21	0.30	0.39	0.29	0.30	0.27	0.10	0.04	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.55	N/A	N/A	N/A	N/A	N/A	N/A	-200	N/A	N/A
Regional (4 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	-227	-265	N/A	N/A

Nomenclature: See separate page